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Extended range prediction of active-break spells of Indian summer monsoon rainfall using an ensemble prediction system in NCEP Climate Forecast System

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ABSTRACT: This study analyses skill of an extended range prediction system to forecast Indian Summer Monsoon Rainfall (ISMR) 3–4 pentads in advance. A series of 45-d forecast integrations starting from 1 May to 29 September at 5-d interval for 7 years from 2001 to 2007 are performed with an ensemble prediction system (EPS) in NCEP Climate Forecast System Version 1 (CFSV1) model. The sensitivity experiments with different amount of perturbation suggest that full tendency perturbation experiment on all basic variables including humidity at all vertical level shows higher dispersion among forecast than other experiments. Spread–error relationship shows that the present EPS system is under-dispersive. The lower bound of predictability is about 10–12 d and upper bound of predictability is found to be 20–25 d for zonal wind at 850 and 200 hPa. The signal-to-noise ratio (SNR) of precipitation (500 hPa geopotential height) reveals that the predictability limit is about 15(18) d over Indian monsoon region. The monsoon zone area averaged precipitation forecasts averaged over 5-d period (pentads) up to 4 pentad lead time are also evaluated and compared with observation. The anomaly correlation coefficients (ACC) reaches zero after pentad 3 (pentad 5) lead for precipitation (dynamical variables). A probabilistic approach is developed from the EPS for extended range forecast applications. The relative operating characteristic (ROC) curves for three categories of precipitation shows that the prediction skill for active and break is slightly higher compared to that of normal category and skillful probabilistic forecasts can be generated for precipitation even beyond pentad 4 lead.

KEY WORDS Indian summer monsoon; extended range prediction; ensemble prediction system

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1. Introduction

Extended range prediction of active-break spells in monsoon over India 2–3 weeks in advance is not only crucial for agriculture planning especially sowing, harvesting, irrigation and water management but it also controls and limits the predictability of seasonal mean. Although deterministic forecasts from dynamical General Circulation Models (GCMs) show continued improvement in skill in short and medium range forecasts, extended range dynamical forecasts over India remains a challenging problem. On the other hand, several studies demonstrated that statistical models show significant skill in predicting precipitation evolution over 2–4 weeks lead time (Webster and Hoyos, 2004; Xavier and Goswami, 2007b; Chattopadhyay *et al.*, 2008). However, these skills in forecasts are on area average precipitation and not on the spatial distributions on which decisions are made.

Intraseasonal oscillation (ISO) is largely responsible for the active-break cycle of Indian Summer Monsoon Rainfall (ISMR) (Lau *et al.*, 1998; Waliser *et al.*, 2003a; Goswami, 2005). The mean spatial structure of rainfall and circulation fields associated with active and break conditions (Ramaswamy, 1962; Ramamurthy, 1969; Krishnamurti and Subrahmanyam, 1982; Krishnamurti *et al.*, 1985; Webster *et al.*, 1998; Krishnan *et al.*, 2000; Annamalai and Sperber, 2005; Goswami, 2005; Joseph *et al.*, 2009) have very large spatial scale extending far beyond the Indian continent. An important characteristics of these intraseasonal spells is the repeated northward propagation of the zonally oriented cloud band from south of the Equator to about 25°N (Yasunari, 1979; Sikka and Gadgil, 1980). Further, the nonlinear relationship between the precipitation and the large-scale circulation indicates that the active-break spells are influenced by air–sea interactions (Sengupta *et al.*, 2001; Fu *et al.*, 2003; Waliser *et al.*, 2003b) and are related to a convectively coupled oscillation consistent with theory (Jiang *et al.*, 2004; Wang, 2005; Wang *et al.*, 2009; Goswami *et al.*, 2011). Therefore, a coupled ocean–atmosphere general circulation model (CGCM) may be essential for predicting active-break spells on extended range.

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Forecast skill in the extended range is largely determined by the model's ability to represent the ISO and associated variabilities. Dynamic extended range forecast (DERF) based on National Centers for Environmental Prediction (NCEP) Medium Range Forecast (MRF) model showed an ISO forecast skill of <7 d (Chen and Alpert, 1990; Jones *et al.*, 2000; Hendon *et al.*, 2000). Waliser *et al.* (2003b) found an extended range predictability for ISO of about 30 d for 30–90 d band pass filtered 200-hPa velocity potential anomalies and about 18 d for precipitation anomalies associated with the ISO in the Goddard Laboratory for Atmospheres GCM. At the same time, regionality of the nature of ISO variability is another major determining factor of extended range prediction skill. Predictability studies by Liess *et al.* (2005) using ensemble hindcast experiments with ECHAM5 GCM found that upper limit for predictability of precipitation and 200 hPa zonal wind is about 27 and 33 d, respectively, over Southeast Asia, while over India, predictability values are limited to about 15 d for both precipitation and dynamic variables. Nevertheless, this potential predictability is considerably higher than for numerical forecasts of typical weather variations, particularly over tropics. Their study provided a hope that useful dynamical forecasts of monsoon active-break cycle may be possible with lead times of more than 2 weeks for precipitation and circulation.

It is also conceivable that for extended prediction skill may also depend on different phases of monsoon ISO's. Xavier and Gowswami (2007a, 2007b) found that transition from break to active is much more chaotic than that from active to break and hence the break phases of monsoon ISO's are intrinsically more predictable than the active phase. Uncertainty in the dynamical extended range prediction mainly arises from the uncertainty in the initial conditions and also from model uncertainty in terms of incomplete representation of the physics of the problem. Ensemble approach is usually used in an attempt to sample some or all these two sources of errors (Palmer *et al.*, 1993; Richardson, 1998; Harrison *et al.*, 1999). The operational ensemble prediction system (EPS) by European Centre for Medium Range Weather Forecasting (ECMWF) produce largest ensemble sets of 51 members in real time (Molteni *et al.*, 1996; Buizza *et al.*, 2007, 2008). Global ensemble system of NCEP produces ensembles of 21 members in their real time operational analysis/forecast cycle (Toth and Kalnay, 1997). Rashid *et al.* (2010) assessed the Madden–Julian Oscillation (MJO) prediction skill of 10-member ensemble of hindcast from POAMA (Australian Bureau of Meteorology coupled ocean–atmosphere seasonal prediction system). They found that MJO can be predicted with POAMA ensemble out to about 21 d as measured by bivariate correlation exceeding 0.5. Vitart and Molteni (2009) have shown that 15 member ensembles of dynamical forecast using ECMWF Variable Resolution Ensemble Prediction System (VarEPS) display some prediction skill during onset phase with a month lead time.

As deterministic predictability limit in NWP range reaches within 10 d, issuing deterministic forecast alone will not add value to forecast in the extended-range time scale beyond 10 d. The main goals of this study are to evaluate the extended range forecast skill of the EPS over India during summer monsoon season based in Climate Forecast System (CFSV1) model of NCEP and to develop probabilistic products from deterministic EPS for extended range forecast applications. In this study, we investigated the sensitivity of different perturbation techniques to the growth of dispersion (spread) as well as signal–noise evolution as a function of forecast lead. Model precipitation forecasts are verified against Global Precipitation Climatology Project (GPCP) precipitation data sets (Adler *et al.*, 2003). Forecast skill of the precipitation is verified for homogeneous zone of core Indian monsoon region called Monsoon Zone of India (MZI) as defined in Rajeevan *et al.* (2008). Precipitation forecast skill for a series of 5-d period (pentads) is also evaluated. This article is organized in the following manner in which perturbation methodology is described in Section 2, sensitivity experiments are discussed in Section 2.1., results and discussions are presented in Section 2.2. followed by summary and future scope in Section 2.3..

2. Data, model and methodology

2.1. Data

As part of National Monsoon Mission (NMM) program, our group carried out a series of 45-d forecast integrations (extended range hindcast runs) using CFS starting from 1 May to 29 September at 5-d interval (total 31 initial conditions during the Boreal summer monsoon season) for 7 years during 2001–2007. The NCEP/NCAR reanalysis-1 (Kalnay *et al.*, 1996), data is used as verification for dynamical and thermodynamical variables (here after ANA) and the forecast precipitation is compared with GPCP ($1^\circ \times 1^\circ$ interpolated to $2.5^\circ \times 2.5^\circ$ to make it comparable with model resolution). For hindcast experiments, the atmospheric initial conditions were obtained from the NCEP/DOE Atmospheric Model Inter-comparison Project (AMIP) II Reanalysis (R2) data (Kanamitsu *et al.*, 2002), and the ocean initial conditions were from the NCEP Global Ocean Data Assimilation (GODAS) (Behringer, 2007). For calculating area averaged precipitation over monsoon zone, gridded precipitation data over Indian landmass (Rajeevan *et al.*, 2006) has been used.

2.2. Model

The atmospheric component of the CFSV1 is the NCEP Global Forecast System (GFS) model (Moorthi *et al.*, 2001). Atmosphere model has an approximate horizontal resolution of 210 km (corresponding to T62 spectral triangular truncations) with 64 levels in the vertical. The GFS model includes parameterization of atmospheric solar radiation (Hou *et al.*, 1996; Hou *et al.*, 2002), boundary layer vertical diffusion (Hong and Pan, 1996), cumulus convection (Hong and Pan, 1998) and gravity wave drag (Kim and Arakawa, 1995).The oceanic component is the GFDL Modular Ocean Model V.3 (MOM3) (Pacanowski and Griffies, 1998). The domain is quasi-global extending from 74°S to 64°N. The zonal resolution is 1°. The meridional resolution is 1/3° between 10°S and 10°N, gradually increasing through the tropics until becoming fixed at 1° poleward of 30°S and 30°N. There are 40 layers in the vertical with 27 layers in the upper 400 m, and the bottom depth is around 4.5 km. More details regarding CFS operational setting at NCEP can be found in Saha *et al.* (2006).

2.3. Perturbation method

The primary motivation of this study is to develop a robust, reliable and flexible EPS in an CGCM frame work, which can be used for real-time operational as well as research purpose. At IITM, as part of National Monsoon Mission (NMM), we carried out the initial testing of the in-house developed EPS system based on the Climate Forecast System (CFS) model version 1. There are various approaches in ensemble perturbation techniques. The EPS of ECMWF evolved from 50 initial perturbations generated using singular-vector technique and an unperturbed control run (Buizza and Palmer, 1995). At NCEP, the ensemble of initial perturbations are generated in a similar way as at ECMWF, but breeding vectors (Toth and Kalnay, 1993) are used instead of singular vectors. At Meteorological Service of Canada (MSC) an Ensemble Kalman Filter (EnKF) with perturbed observations and different combinations of parameterization schemes is used to generate ensemble of initial conditions for their medium range prediction (Houtekamer *et al.*, 1996, 2005). Buizza *et al.* (2008) classified the different ensemble generation strategies into three groups. Although there are several approaches to generate ensembles of different initial conditions, we use an approach which is similar to the 'complex-and-same-model environment group' as classified in Buizza *et al.* (2008).

An ensemble of 10 perturbed atmospheric initial conditions were developed in addition to one actual initial condition. Each ensemble member was generated by slightly perturbing the initial atmospheric conditions with a random matrix (random number at each grid point) generated from a random seed. In order to make the perturbation size consistent with analysis variance of each variable, the amplitude of perturbations are adjusted to ensure sufficient spread in the forecast fields and to also ensure that amplitude of perturbation varies in accordance with the uncertainty in the analysis. In order to make the ensemble mean to be centered around the unperturbed analysis, fraction of the tendency of different model variables are added to or subtracted from the unperturbed analysis with random perturbation between −1 and +1 times the tendency so that the perturbations follows a Gaussian distribution. We perturb the wind, temperature and moisture fields and the amplitude of perturbation for all variables are consistent with the magnitude of variance of each variable at a given vertical level. The procedure for generating initial perturbations is further described below.

The variable X after perturbation can be expressed as,

$$X'_{x,y,z,t} = X_{x,y,z,t} - n\left[r\,\Delta X_{x,y,z,t}\right] \quad (1))$$

The tendency is calculated as, $\Delta X = X_{x,y,z,t} - X_{x,y,z,t-1}$ where $X_{x,y,z,t}$ is the actual analysis at time t and $X_{x,y,z,t-1}$ is the analysis valid for the previous day. The dimension x changes in longitude ($x = 1,2,3,....144$), y changes in latitude ($y = 1,2,3\ldots73$) and z changes in vertical levels ($z = 1,2,3\ldots28$). The number 'r' is taken from the random matrix and it lies between −1 and 1 and 'n' is a tuning factor. Where value of n is such that $0 < n \leq 1$ and $n = 1$ corresponds to 100% tendency perturbation. Main advantage of this system is that, it has a potential to generate infinite number of ensembles and also the amplitude of perturbations can be adjusted by changing the tuning factor.

It is well known that the skill of an EPS essentially depends on the ensemble size (Richardson, 2001; Reynolds *et al.*, 2011). However, running large ensemble members in real-time basis require huge computational power and most of the operational centres limit the ensemble size to few tens. In his analysis, number of perturbed ensemble members are limited to 10. In the present formulation, there is option to select the perturbation variable, where X can be any variable, u-component of wind (u), v-component of wind (v), temperature (t) and humidity (q). This approach also helps to study the individual impact of perturbing each variable on the final forecasts.

2.4. Sensitivity testing

Two experiments were conducted to evaluate the impact of tuning factor n on the final forecast fields. Many more integrations are required to evaluate the impact of the tuning factor ranging between 0 and 1, this however, is too much time consuming and requires lot of compute power and storage space. A practical trade-off is to examine the impact of two extreme cases with n = 1 corresponding to high perturbation (HIGHPER) and n = 0.05 corresponding to low perturbation (LOWPER) such that these two cases give an idea about the forecast characteristics at maximum and minimum perturbation. In these experiments, we perturbed all basic variables (u, v, t and q). The precipitation forecast is primarily determined by the temperature and humidity and many studies showed the importance of moisture analysis on precipitation forecast (Puri and Miller, 1990; Krishnamurti *et al.*, 1991). The sensitivity of initial uncertainty in the humidity analysis on precipitation forecast is evaluated by conducting one more additional experiment. This experiment (HIGHPER_NQ) is same as HIGHPER but we perturbed all other variables except humidity. In this experiment, we compared the impact of dry and moist perturbations on the growth of spread.

Figure 1(a)–(c) shows the actual 850 hPa zonal wind (u850) analysis, perturbed initial condition from HIGHPER and perturbed initial condition from LOWPER

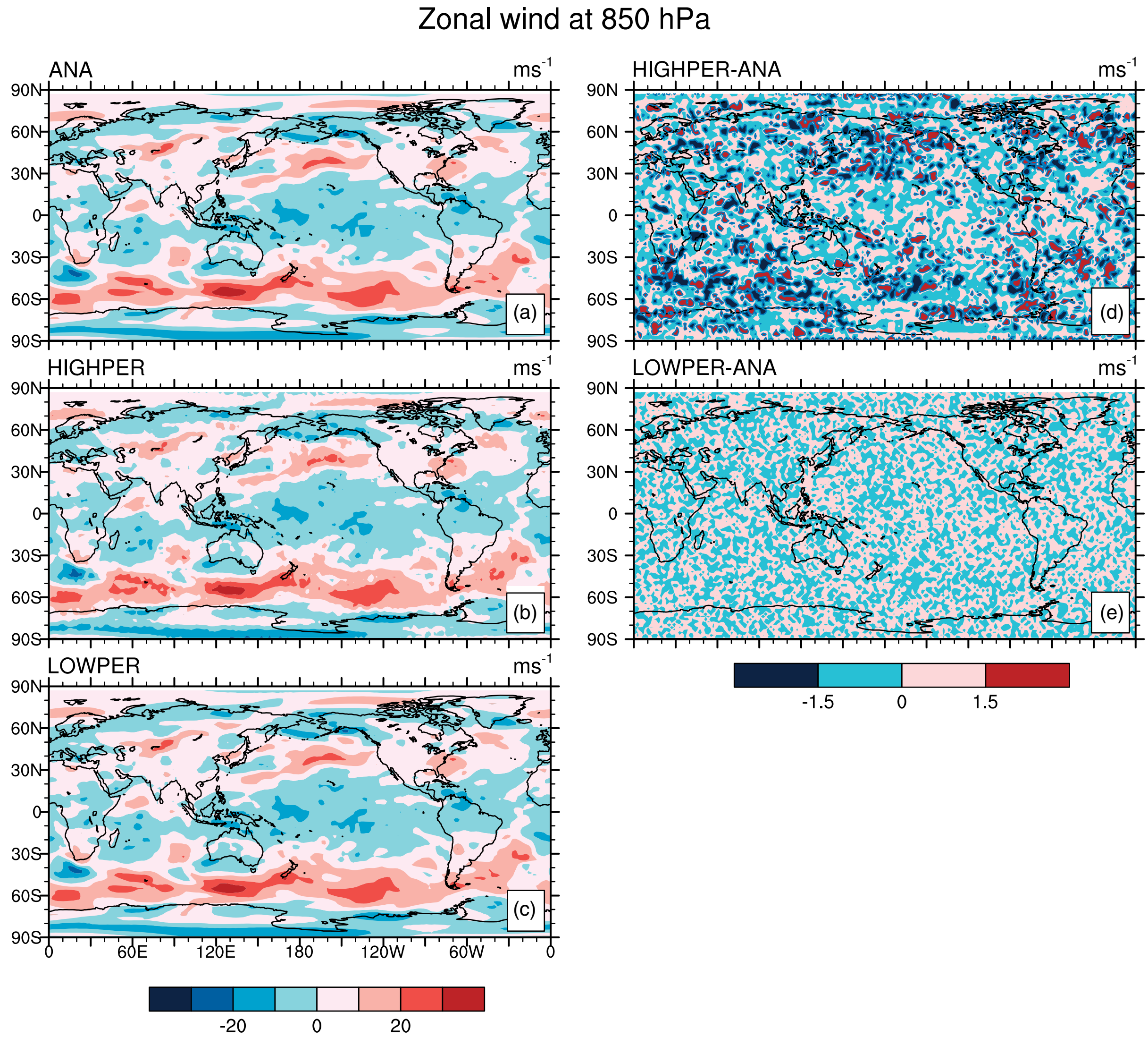

Figure 1. Left panel: Zonal wind (m s^{-1}) at 850 hPa on 1 May 2001 for (a) ANA, (b) HIGHPER, and (c) LOWPER. Right panel: Wind difference (m s^{-1}) at 850 hPa for (d) HIGHPER-ANA and (e) LOWPER-ANA.

experiments, respectively, valid for the same analysis time. It is noted that there is not much difference between the perturbed and actual analysis fields. The difference between the perturbed fields and analysis is also plotted along the right panel. Figure 1(d) and (e) shows the difference between HIGHPER and analysis (HIGHPER-ANA) and LOWPER and analysis (LOWPER-ANA), respectively, for u850 wind. Note that the differences are rather small for the most part except some places where large climatological variance of daily analysis exists. There exists noticeable difference between the perturbed and actual analysis (see right panel of Figure 1) around extra tropics. This is due to the fact that analysis variance of the wind and temperature fields are higher in the extra tropics and that we adopted a perturbation strategy in which the magnitude of perturbation is proportional to the tendency of each variables perturbed.

The impact of perturbation factor on forecast field is assessed by plotting the time series of precipitation over core monsoon zone of India (hereafter known as MZI) covering central India for 45-d forecast lead. Two distinct initial conditions have been selected here to evaluate the growth of spread under two different atmospheric base sate. Figure 2(a) and (b) shows time series of the area averaged precipitation forecasts over MZI for 45 d starting from 1 May 2001 and 24 August 2001. The left panel on Figure 2 represents the growth of spread from forecast starting from an initial atmospheric condition much before the onset of monsoon. The right panel on Figure 2 presents the forecast started from an entirely different atmospheric state in which monsoon circulations were already established. The solid lines represents ensemble mean and control forecast and spread among members is also included as vertical bars. Obviously, the growth of spread and hence the predictability varies with different atmospheric initial conditions. For 1 May initial condition, the dispersion started growing during early forecast hours at a lead of 7 d, whereas from 24 August initial condition, the dispersion in forecast stared increased from a forecast lead at about 12 d later.

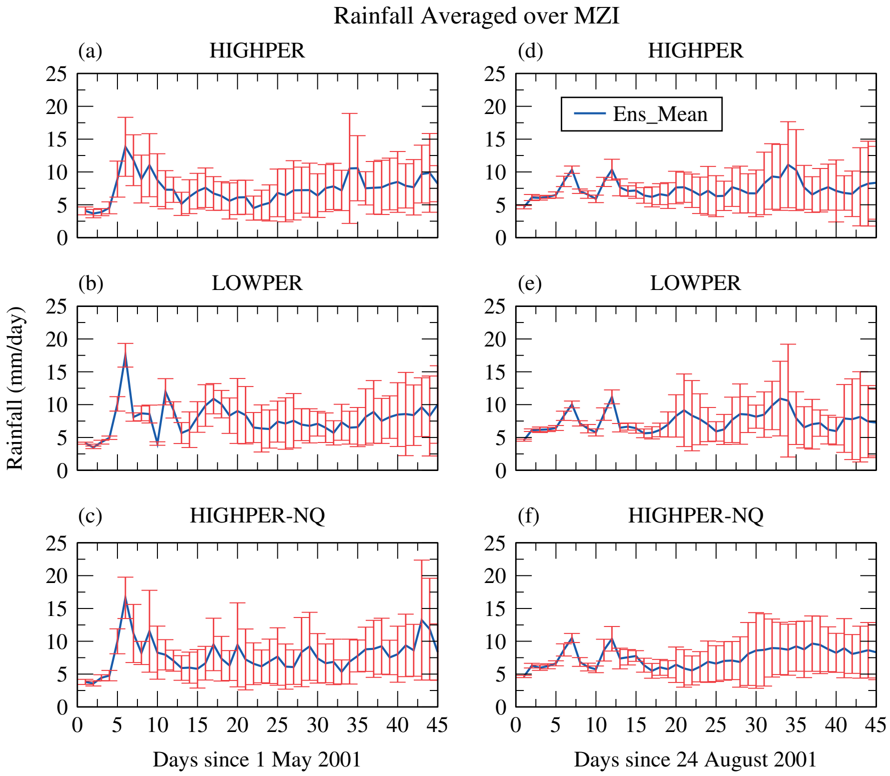

Figure 2. Left panel: Time series of rainfall (mm d^{-1}) evolution starting from 1 May 2001 averaged over MZI for (a) HIGHPER, (b) LOWPER and (c) HIGHPERNQ. Right panel is same as left panel but forecasts starts from 24 August 2001, the vertical bar represents the spread among members.

More study is required in this direction to quantify the growth of spread under different atmospheric base state. It is also evident that the forecast ensemble members are distributed around the control (CTRL) experiment and dispersion is slightly higher for the HIGHPER than for the LOWPER (Figure 2(a) and (b)). Apparently, higher forecast scattering for HIGHPER benefit from the slightly higher analysis difference compared to LOWPER experiment (Figure 1). It is also noted that the dispersion of forecast fields of HIGHPER_NQ is slightly lower than HIGHPER experiment. Thus, perturbation of all variables along with humidity is required to produce large dispersion in forecast. As the dispersion between the analysis field and forecast field for HIGHPER is slightly higher than LOWPER, we fixed HIGHPER for subsequent hindcast experiments. Further, we carried out all hindcast experiments with HIGHPER configuration in which perturbation is added to all variables. Results presented in this article mostly refer to the 850 and 200 hPa zonal and meridional wind, 500 hPa geopotential height and precipitation.

3. Results and discussion

3.1. Simulation of mean monsoon precipitation

For a reliable dynamical modelling system to be used for monsoon forecasting and diagnostics, it is believed that the main features and general characteristics of mean monsoon need to be well simulated. Ability of the EPS in simulating the monsoon precipitation over India is assessed by plotting the area averaged precipitation forecasts over MZI region. The climatology of ensemble mean of the 45-d forecasts starting from each start date at 5-d interval during 2001–2007 very well captures the seasonal cycle of monsoon precipitation (Figure 3). Observed climatology from GPCP during monsoon season is also included in the figure for comparison.

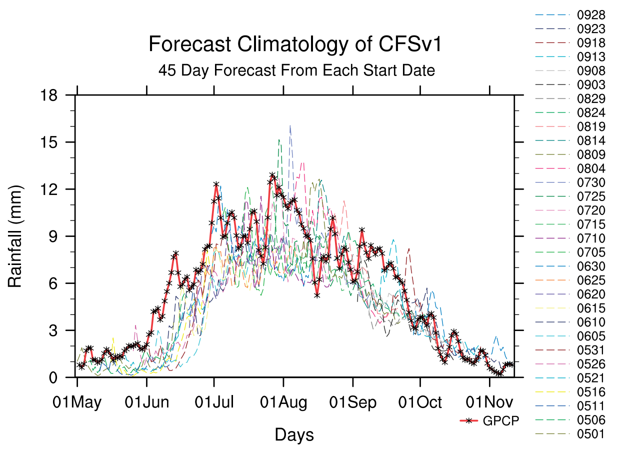

Figure 3. Seasonal cycle of precipitation climatology (mm day^{-1}) from GPCP (solid line) and CFSv1 forecasts climatology starting from each start date for 45-d lead time.

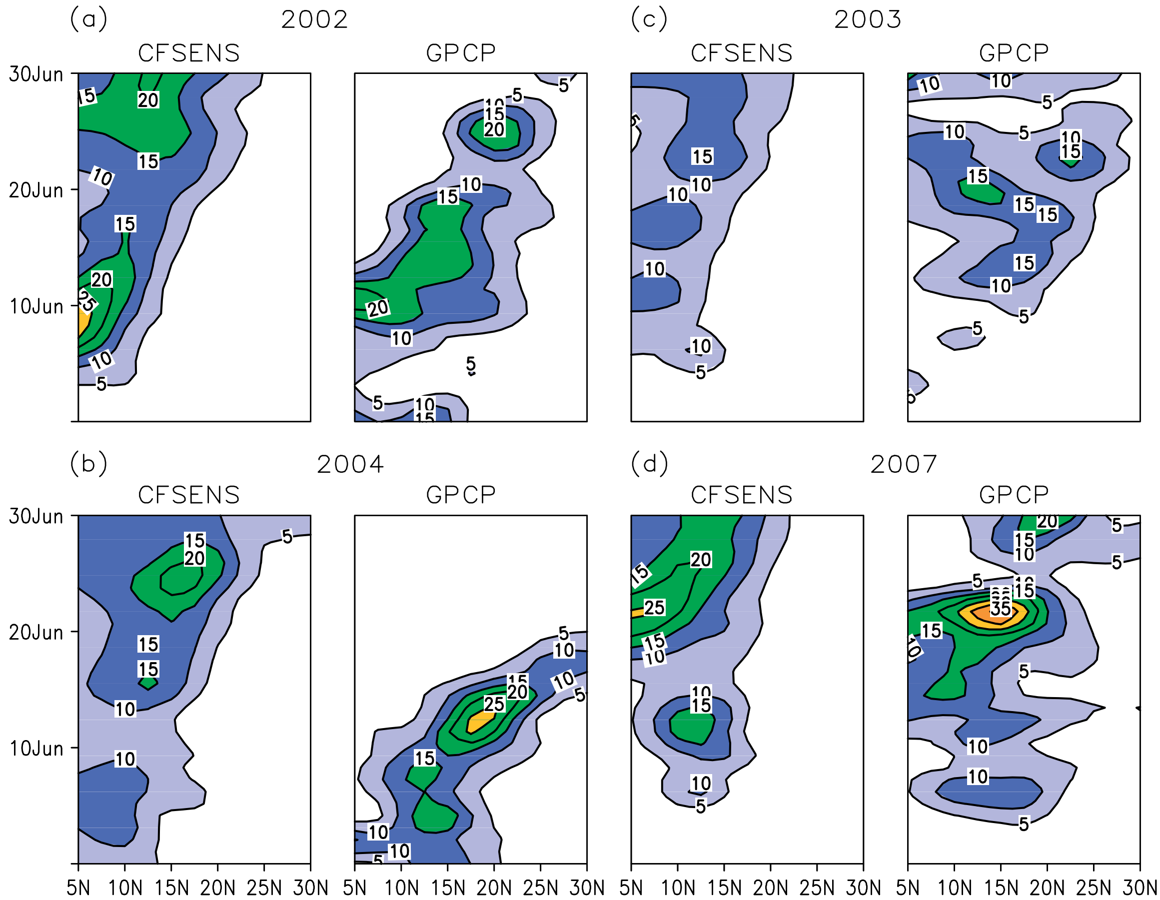

Figure 4. Hovmoeller diagram of daily rainfall (mm d^{-1}) averaged between 70 and 85°E starting from 31 May for (a) 2002, (b) 2004, (c) 2003 and (d) 2007 from the EPS (left) and GPCP (right).

A unique characteristic of the southwest Indian summer monsoon precipitation is its northward propagation from near equatorial Indian Ocean to the Indian subcontinent (Yasunari, 1980). Generally, the propagation characteristics of the monsoon precipitation are evaluated by plotting hovmuller diagram by averaging the precipitation from 70 to 80°E. Model also exhibits significant inter-annual variability in the prediction skill, which is discussed later. There is noticeable year to year variability in the model ability to capture northward propagation features. To see the model's ability in simulating northward propagation, we have selected four typical years in which All India Summer Monsoon Rainfall (AISMR) is above normal during 2 years (2003 and 2007) and below normal during other 2 years (2002 and 2004). Figure 4(a) and (b) shows the hovmoller plots of precipitation during June 2002 and 2004. Left panel in each figure shows the ensemble mean forecast from CFS and observations from GPCP for corresponding periods are shown on right side of each panel plot. During onset phase of 2002 monsoon, model able to capture the second pulse of rainband and which is more or less matching with GPCP observations, but propagation speed is slightly slower in model simulation compared to observation. However, during 2004, model did not simulate the northward propagation in terms of intensity and propagation speed. As seen from GPCP observations, the northward propagation started early in June and reaches 30°N latitude very fast, whereas model simulations showing some false alarms between 20 and 30 June. During 2003 and 2007, model performed better in simulating the northward propagating rainbands. One interesting thing common to all these selected years is that, model simulated precipitation never propagates beyond 25°N latitude (Figure 4(c) and (d)). It is also to be noted here that ensemble forecast has a general tendency to underestimate the precipitation and propagates slightly slower.

3.2. Spread, error and predictability

Spread is a measure of uncertainty in the ensemble forecasts and is simply defined as the standard deviation of ensemble members from ensemble mean. Growth of spread as a function of forecast lead time determines the forecast uncertainty (and reliability) for the EPS system. Large spread in the ensemble forecast for particular lead indicates that the uncertainty in forecast is more and hence the confidence is less. One of the desirable properties of the EPS is assessed by examining the growth of spread and RMSE as a function of forecast lead time. The RMSE is calculated using NCEP reanalysis data sets. The spread indicates growth of small initial errors and also gives an idea about the potential limit

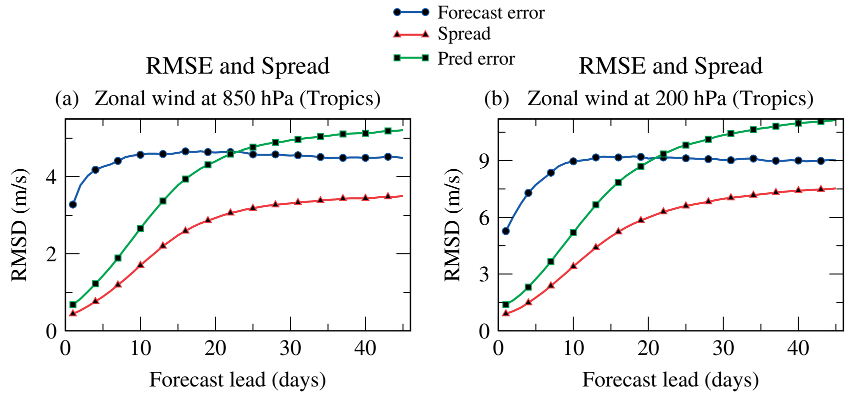

Figure 5. Spread (dot line), RMSE (dash line) and predictability error (solid line) in zonal wind (m s^{-1}) for the tropics (20°S to 20°N) at (a) 850 hPa and (b) 200 hPa.

on predictability. However, RMSE on the other hand shows the growth of actual initial errors or forecast error. The RMSE is also considered as the lower limit of practical predictability. Predictability error is also computed as the RMS deviation between the ensemble forecasts by assuming any one member as perfect model. Thus, RMS spread is constructed as the average of the sum of the RMSE's between every EPS member and all other members. This gives an idea about the upper bound of predictability under a perfect model assumption. Figure 5(a) and (b) show the ensemble mean RMSE (blue), predictability error (green) and spread (red) for zonal wind over tropics at 850 and 200 hPa. Obviously, Figure 5 indicates that for both u850 and u200, the spread is always less than RMSE of ensemble mean. In other words, ensemble spread is under dispersive or smaller than the RMSE of ensemble mean forecast which is a desirable or pre-requisite property for a reliable EPS. Because the spread is smaller than RMSE, it is likely that the present EPS system does not capture all possible sources of error from internal dynamics (Vialard *et al.*, 2005). This is the general behaviour of present day EPS, that there is lack of spread around ensemble mean or system is under-dispersive (Buizza *et al.*, 1999, 2008; Bengtsson *et al.*, 2008).

Our extended range experiment from EPS shows that the spread and RMS values increases coherently during short to medium range time scale (up to 7–8 d), then RSME is almost constant between 10 and 20 d, while spread continues to increases till it reaches saturation near forecast lead of 20–22 d (see Figure 5). Initially, spread and predictability errors were close to each other and then they diverge with forecast lead and predictability error approaches forecast error at about 22 d. At larger leads, spread is considerably lower than both forecast and predictability errors and this may be due to the model bias and ensemble mean forecast tend to saturate towards a basic model climatology. A noticeable feature from this figure is that the RMSE of the ensemble mean saturates faster than predictability error and ensemble spread. As can be seen from Figure 5, the initial error is very large and it is already in the nonlinear region of error growth and hence saturates fast. Reynolds *et al.* (2011) reported similar spread and RMSE evolution in their medium range forecast experiment with a global EPS. However, their study was limited to forecast lead time of a week (7–9 d). In a seasonal ensemble prediction experiments Vialard *et al.* (2005) found that spread of forecasts from different ensemble generation techniques were smaller than RMSE and concluded that model error was the main contributor affecting not only the ensemble mean but also the ensemble spread. Reynolds *et al.* (2008) tried to identify the possible sources of errors attributed to initial condition and model uncertainties. They found that initial perturbations accounted for most of the improvements during shorter forecast leads and improvements on longer forecast leads are achieved by including a stochastic kinetic energy backscattering scheme for generating initial perturbations.

It is also important to investigate the relationship between noise, defined as standard deviation of ensembles from the ensemble mean, and signal, defined as the standard deviation of ensemble mean forecasts valid at the same forecast lead time. Figures 6 and 7 depict the pentad spread, signal and RMSE of u200 and v200. Obviously, the signal and RMSE of the EPS system has a strong regional dependence which is determined by the variance of field analysed. For 200 hPa zonal wind, spatial pattern of signal, noise and RMSE averaged over 5 d (pentads) suggests that there exists large spatial variability in the relationship between signal, noise and RMSE (Figure 6). Although noise is distributed almost homogeneously over tropical Indian Ocean, Indian subcontinent and maritime continent region (left panel of Figure 6), distribution of signal shows larger spatial variability. Tropical Indian Ocean region and parts of Western Pacific have higher signal than other region. Signal remains more or less same through all forecast leads except, Western north Pacific, China Sea and southern Indian Ocean, where it decreases with forecast lead. Ideally, signal should remain same and noise take over signal as forecast lead increases. Climatology also varies with forecast lead (figure not shown) and signal for each forecast lead might be depended on the climatology at corresponding

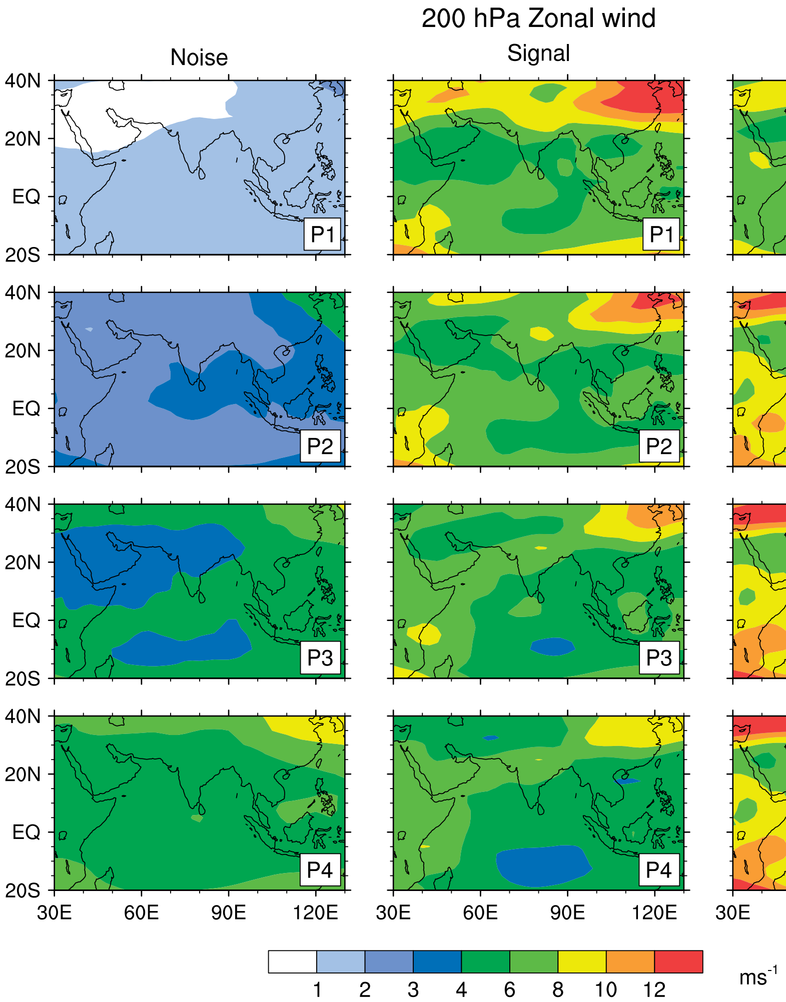

Figure 6. Pentad average noise (left panel), signal (middle panel) and RMSE (right panel) for 200 hPa zonal wind (m s^{-1}) up to four pentad lead.

forecast lead and thence the model bias. At pentad 4 lead, higher signal is confined to tropical Indian Ocean, southern peninsular India and tropical western Pacific. Perhaps, this region corresponds to the large inter-annual variability in the zonal wind associated with seasonal circulation pattern. A general correspondence between spread and RMSE is also noticed in spatial distribution. Unlike spread, RMSE distribution is not smooth, which display some local pockets of large variability. Large RMSE values are also found over the Bay of Bengal, Arabian Sea, Western pacific region and south Indian Ocean between 5 and 10°S. These regions are found to exhibit large intraseasonal variability in both zonal wind and precipitation (Teng and Wang, 2003). The middle panel in Figure 6 shows that most of the predictable signal is seen over this tropical belt, especially over tropical western Pacific and Indian Ocean region.

The noise structure of meridional wind at 200 hPa (v200) also exhibits almost similar spatial pattern as that of u200 (Figure 7). Signal decreases more rapidly with forecast lead compared to rate of change of signal for zonal wind (see Figure 6). Signal found to decrease over most of western Arabian Sea off the coast of Arabia and adjoining land area and over western north Pacific region. On the other hand, noise increases with forecast lead over these two regions. Especially, over the tropical region, the noise is less than RMSE even beyond pentad 4 lead. However, at 4 pentad lead, outside tropical Indian and Western Pacific region, the spatial pattern of noise is similar to the RMSE. Unlike for u200, signal and noise of v200 becomes comparable in magnitude at 4 pentad lead over entire tropical belt. The coherent evolution of the spatial pattern of noise and RMSE suggests that the EPS system is reliable for lead times greater than 20 d for dynamic variables like zonal and meridional wind. However, the amplitude of RMSE is much larger than the spread of the ensemble. The spatial dependence on noise, signal and RMSE relationship shown in Figures 6 and 7 suggests that noise, signal and RMSE vary according to the observed variance of each field. This analysis shows that the location of maxima in spread of the ensemble is reasonably well coincides with the location of maxima in RMSE and provides a prior information about the unpredictability in the forecast.

Growth of spread between the ensemble members also gives an idea about the predictability, and provides

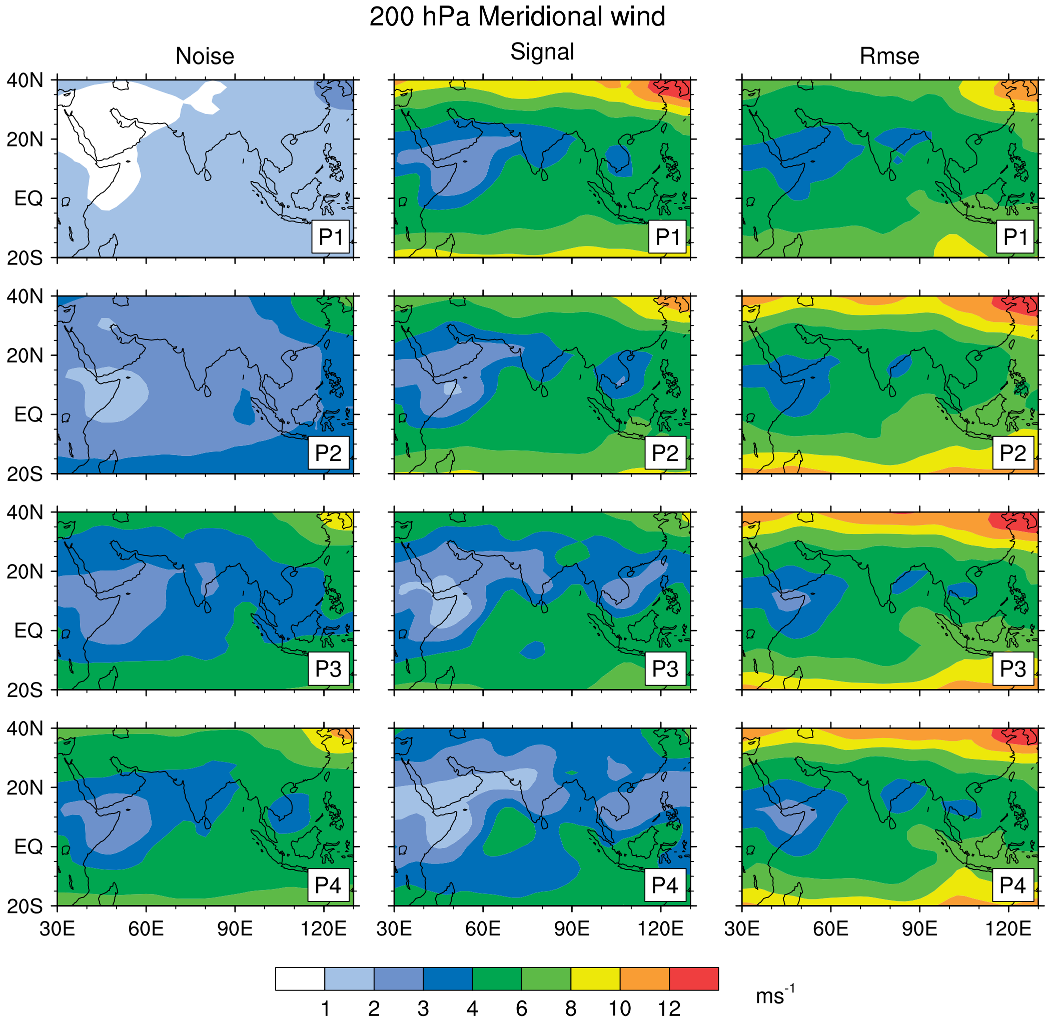

Figure 7. Pentad average noise (left panel), signal (middle panel) and RMSE (right panel) for 200 hPa meridional wind (m s^{-1}).

some guidance for the ensemble size and choice of initial perturbations to provide reliable forecast. For the assessment of predictability, further analysis is carried out to quantify the practical predictability limit of the present EPS during Indian summer monsoon season. Figure 8 depicts spatial patterns of SNR for precipitation at pentad intervals up to 4 pentad lead. It is evident from the figure that the signal is higher than noise up to third pentad and thereafter the noise becomes comparable, or greater in magnitude than signal or in other words, SNR becomes less than or equal to one.

A succinct measure of predictability limit for the EPS can be evaluated by computing the SNR. The SNR value of one is generally considered as a limit of useful predictability. Right panel in Figure 9 shows the lead time at which the SNR become less than one. We further consider the SNR over core Indian monsoon region (MZI). The time series of signal and noise for precipitation over MZI in Figure 9(a) shows that upper limit on predictability is about 15 d. However, for 500 hPa geopotential height (z500) (Figure 9(b)), predictability increased to 18 d. Large signal compared to noise suggest that most of the variability is predictable up to forecast lead of 15 d for precipitation and 18–20 d for 500 hPa geopotential field. This predictability limit is considerably higher than for numerical prediction of typical weather and synoptic events and useful forecasts may be possible up to a lead time of three pentads for precipitation and up to four pentads for other dynamical variables.

Figure 9(c) and (d) shows the spatial plots of forecast lead at which SNR becomes less than one for v200 and z500. The lead times in Figure 9 provide additional information regarding the spatial variability in the predictability limit. Areas with strong signal (Figure 8) not necessarily coincide with areas of higher predictability lead time. Central and western Arabian Sea exhibit strong predictability limit of more than 25 d for v200 and z500. Over most of the Indian region, the predictability reaches beyond 15 d for both v200 and z500. However, as discussed earlier in Figure 7, there exists a region of lesser predictability just northwest of the maximum predictability zone and also some local pockets of low predictability

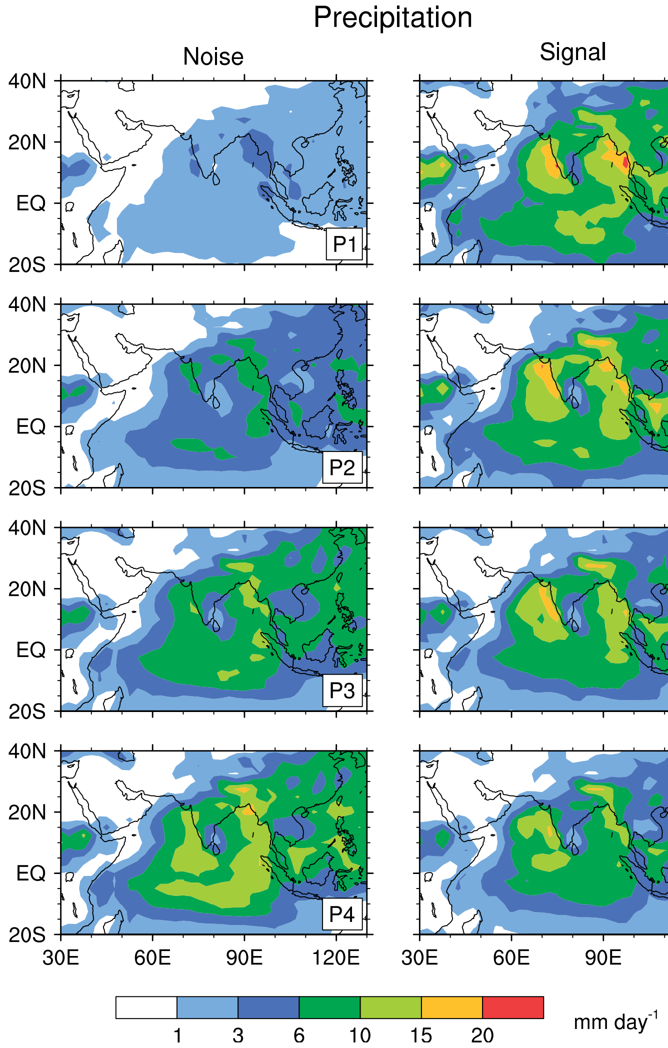

Figure 8. Pentad average noise (left panel) and signal (right panel) for precipitation in millimetre.

values over northern west Pacific. Although model estimates of SNR ratio shows potential predictability of features beyond 15 d, it is very difficult to translate the information to real-time forecasting scenario, where accurate estimation of the level of noise in real world is difficult (Vialard *et al.*, 2005). Using 30- to 90-d band pass filtered data from ensembles with random perturbation, Waliser *et al.* (2003b) found that dynamical predictability limit of ISO's associated with Asian summer monsoon is about 15 d for precipitation and 25 d for 200 hPa velocity potential. Later Liess *et al.* (2005) reported that this band pass filtering leads to overestimation of predictability compared to projecting the forecast on first four EOF's.

3.3. Forecast verification and prediction skill

Forecast verification is an essential component of any EPS. This section is devoted exclusively for the various verification measures to quantify the extended range prediction skill of the present EPS. Although we developed a global EPS, our forecast skill analysis mostly focused over Indian region at pentad intervals. The pentad prediction skill may be considered as total intraseasonal variability (ISV) prediction skill and is a more rigorous way of evaluating the model's hindcast skill. The verification skill scores are shown for ensemble mean forecast as deterministic and also considering the individual member forecast for probabilistic skill measures. As deterministic predictability limit in NWP range reaches within 10 d, issuing deterministic forecast alone will not add value to forecast in the extended-range time scale beyond 10 d. As an example, the deterministic and probabilistic forecast of precipitation over MZI region for three pentad lead from 2001 to 2007 has been shown in Figure 10. The categories of forecast are defined in such a way that if the area averaged precipitation is greater than 40% of its long-term mean then it is considered as active period, if it is less than −40% then it is break period, and otherwise a normal period. These categories are defined independently for EPS forecast and observations. It is clear that the probability of three categories favoured the ensemble mean precipitation anomaly. Small ensemble size of 11 members might also have an impact on the probability forecast and a wide range of probability

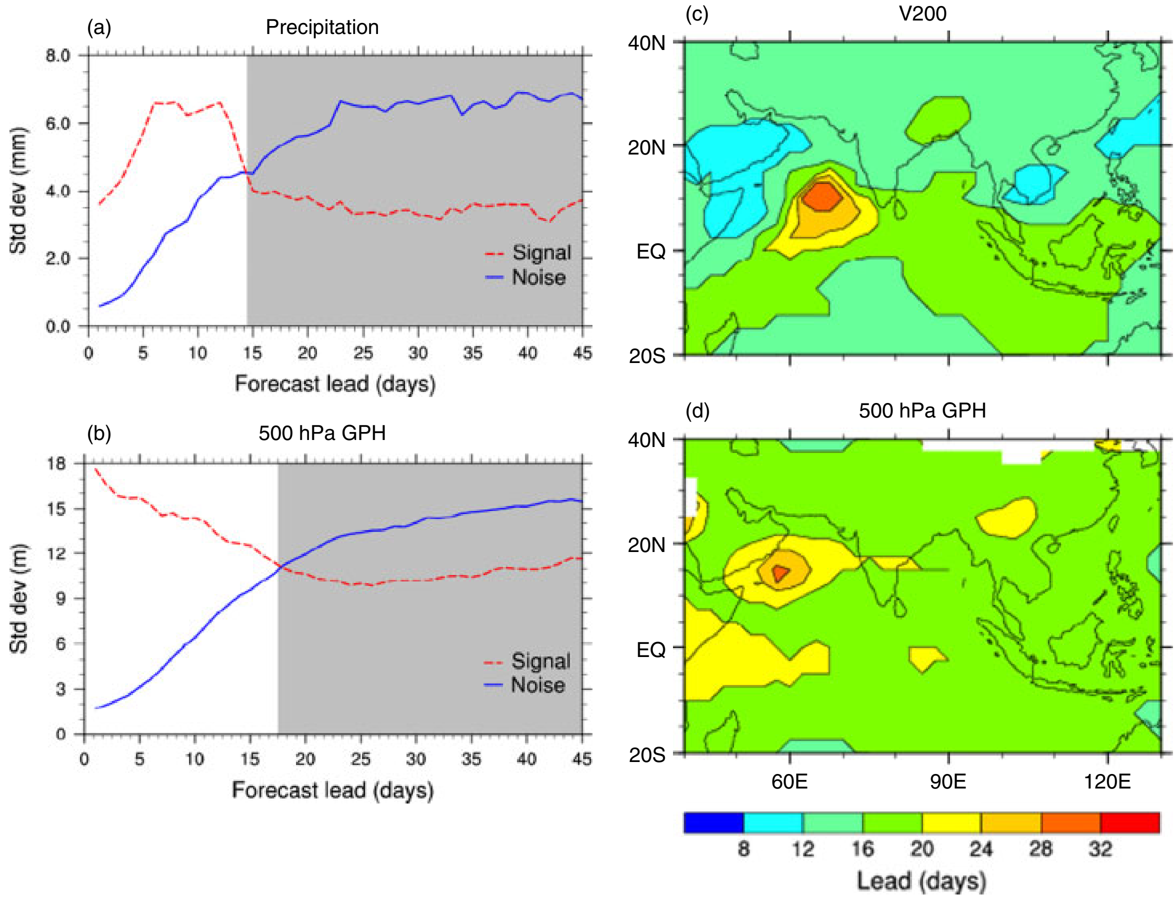

Figure 9. Area averaged signal and noise of (a) precipitation (mm d^{-1}) over MZI for 45-d forecast lead, (b) z500, the forecast lead in days at which noise crosses signal for (c) v200 (m s^{-1}) and (d) z500.

values can be achieved only by increasing the ensemble size.

Anomaly correlation coefficient (ACC), a common measure of the deterministic forecast skill has been computed over Indian region. Model anomaly is calculated from model forecast climatology corresponding to each start date (e.g. 1 May, 6 May, 11 May, etc.). Anomaly correlation coefficient of precipitation over MZI and anomaly pattern correlation coefficient of other dynamical variables (u850, u200, v850, v200) averaged over 60–110°E and 10–40°N is presented in Figure 11(a)–(f). The ACC for precipitation is calculated from GPCP observations interpolated to the model grid (resolution) and NCEP reanalysis data sets are used to verify other dynamical variables.

At first pentad lead forecast, ACC of precipitation is near 0.4 and decreases sharply and reaches zero after third pentad. This is consistent with the predictability estimates discussed in the previous section (Figure 9) based on the SNR, in that skillful forecasts for precipitation may be given up to third pentad (15 d) lead. As far as dynamical variables are concerned, ACC values are slightly higher (around 0.5–0.6) at first pentad lead and decreases to zero around fifth pentad lead forecasts. The ACC values of dynamic variables greater than zero up to fifth pentad lead also supports that there exists predictable signal in dynamical variables even beyond fifth pentad as was discussed earlier.

Figure 11(b) shows the ACC of precipitation over MZI during each individual year. It is evident from the Figure 11 that current EPS also shows inter-annual variability in the precipitation forecast skill. At pentad 1 lead, ACC values of more than 0.35 is obtained for all years except 2003 and 2006. The same is true for pentads 2 and 3 lead forecasts where ACC values become negative. At pentad 4 lead, ACC of precipitation becomes negative except for 2001 and 2002. Forecast skill in terms of ACC values suggests that 2001 and 2002 shows large prediction skill even up to 4 pentad lead. Unlike for precipitation, dynamical variables do not exhibits large inter-annual variability (see Figure 11(c)–(f)). For dynamical variables, large ACC values of about 0.5–0.6 at pentad 1 lead persistently decreases to a value of about 0.1 at fourth pentad lead

An assessment of value-added forecast skill is also developed based on the dynamical ensemble forecasts. Categorical forecast skill score is computed by using discrete forecasts grouped into different classes based on a threshold value. Different possibilities are thus obtained from a range of several threshold values from a 2 × 2 contingency table in which forecast-observation pairs are classified into four different groups (Wilks, 1995).

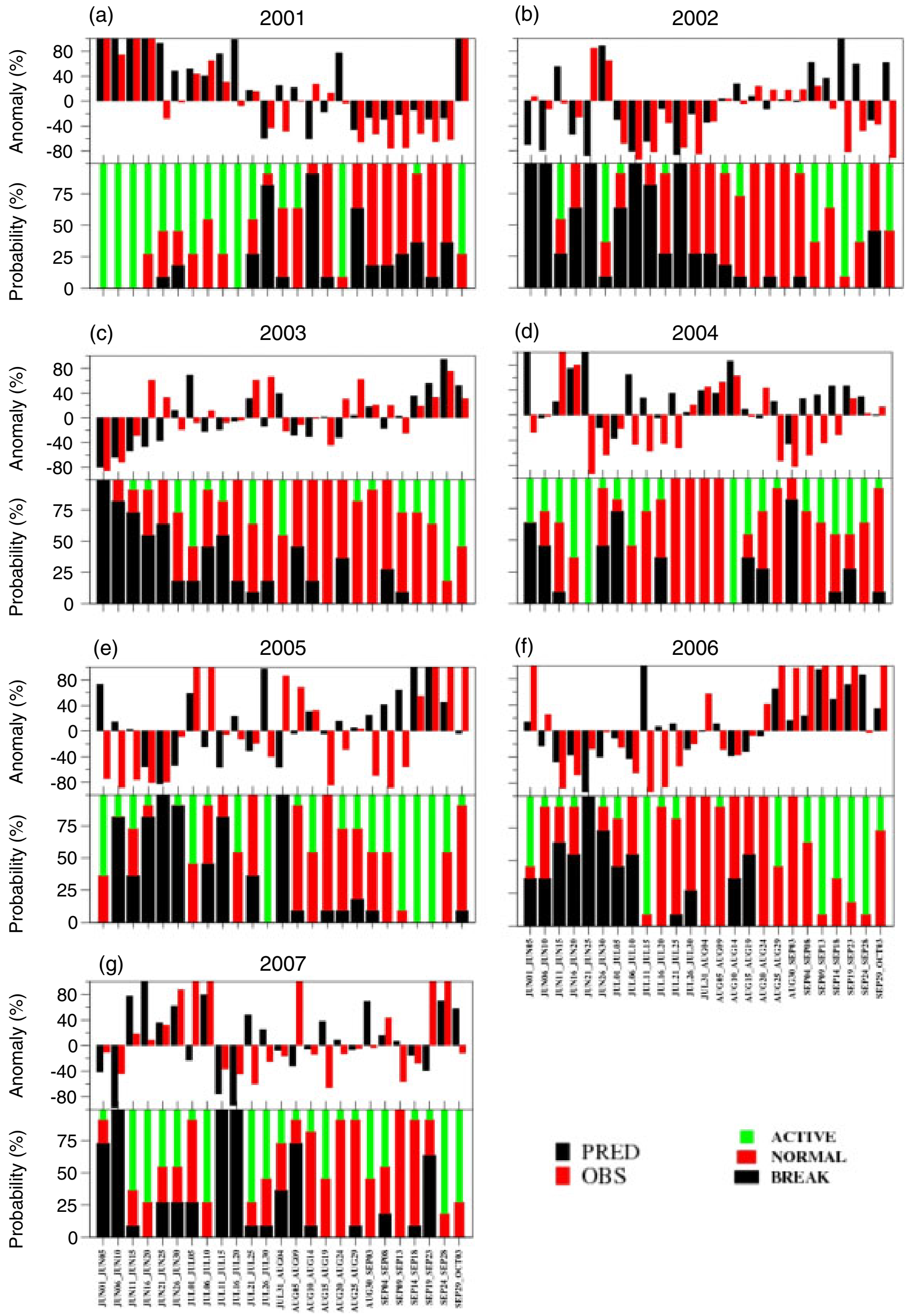

Figure 10. Third pentad lead forecast for years 2001–2007 over MZI. Top panel in each figure contains ensemble mean forecast and bottom panel shows the percentage probabilities in three categories active, normal and break.

Traditional precipitation forecast verification scores such as false alarm rate (FAR) and hit rate is used to analyse categorical forecast skill over MZI region. Here, we choose relative operating characteristic curves (ROC) to assess the forecast skill by defining the dichotomous event such as occurrence or non occurrence of the precipitation above or below certain thresholds. Basically, ROC combines the hit rate and false alarm rate. The ROC has been generated from 11 member ensemble forecast. The ROC has been gained widespread acceptance as a tool for probabilistic as well as ensemble forecasts verification (Mason and Graham, 1999; Hamill *et al.*, 2000; Hamill and Juras, 2006).

In the present analysis, ROC for three observed categories is evaluated. The three observed categories are defined in such a way that break corresponding to percentage departure of precipitation over MZI below −40%, normal ranges between −40% and +40% and an active range above +40. The 11 member ensembles prediction of area averaged percentage departure is sorted from lowest to highest and which is then converted to yes/no forecast by comparing it with the observation and

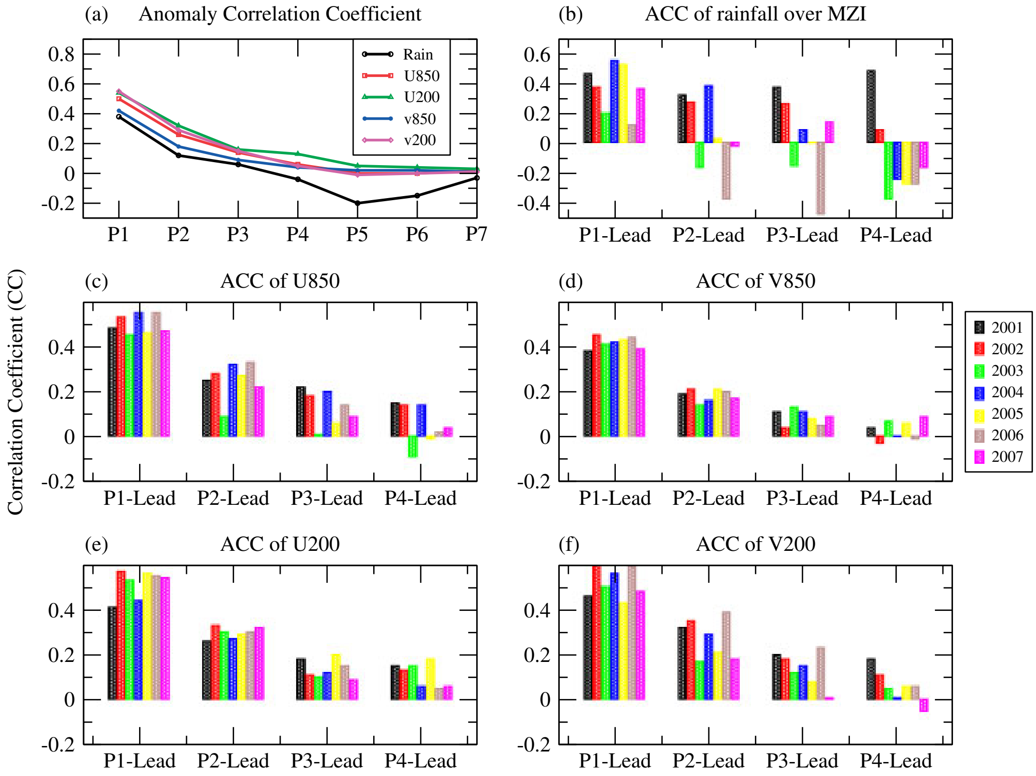

Figure 11. (a) Pentad ACC in precipitation averaged over MZI, zonal (u850 and u200) and meridional wind (v850 and v200) over Indian region for all years, (b) ACC of precipitation during each individual year (2001–2007) for up to four pentad lead, (c–f) is same as (b) but for pattern ACC values of u850, v850, u200 and v200.

assigned a binary value (1 for correct forecast case and 0 for not forecasted). Here, three categories of observations are also converted to binary value of 1 or 0 according to the percentage departure of precipitation. The area under the ROC curve can be used for the calculation of a probabilistic skill score. Then separate 2 × 2 contingency tables are calculated for each sorted ensemble member with different probabilities. In this study, the forecast distribution from ensemble system is arranged into 10% wide probability range bins so that total 10 probability classes are obtained. The curve against hit rate and false alarm rate (ROC) for all 10 probability thresholds for first to fourth pentad lead forecast is plotted in Figure 12.

The area under the ROC is evaluated for three categories namely, break, normal and active cases. It is evident from the Figure 12(a) that at pentad 1 lead, the EPS has higher skill in predicting break followed by active and then normal category. It is also to be noted that the skill in predicting break for different probabilities are almost same and in the figure different probabilities are clustered around. Skill of predicting active and normal at different probabilities shows a range of values with slightly less skill compared to break prediction. For pentad 2 lead, skill of predicting both break and active is similar for higher probability values and at probability values <50%, skill for predicting active becomes higher than that for break (see Figure 12(b)). One interesting aspect to note is that curve for normal more or less coincides with the diagonal and in other words, it has an equal chance of hit and false alarm rate and considered as little or no skill at all probability ranges. For pentad 3 lead, Figure 12(c) shows that active category has slightly higher skill than break and normal categories for all probability values. Obviously, there is relatively little skill at pentad 4 forecast for all three categories (see Figure 12(d)). However, the EPS shows that the hit rate is slightly higher than false alarm rate and probably there exists some skill in probabilistic prediction from ensemble system. Of course, the skill is likely to probably depend on the ensemble size (Kumar *et al.*, 2001). Unlike as expected from the ACC skill analysis discussed in this section, ROC analysis shows that skillful probabilistic forecasts may be produced from the EPS system even beyond pentad 4 lead. In general, the forecast skill deteriorates with the forecast lead.

4. Summary and conclusions

An EPS has been developed in a CGCM frame work using CFSv1 model. The present EPS system falls under the category of complex-but-same model environment group as classified by Buizza *et al.* (2008). The present EPS developed in CFSv1 model has the capability of generating as many number of ensembles and also parameters can be tuned or scaled to test the impact or sensitivity of different ensemble generation strategies.

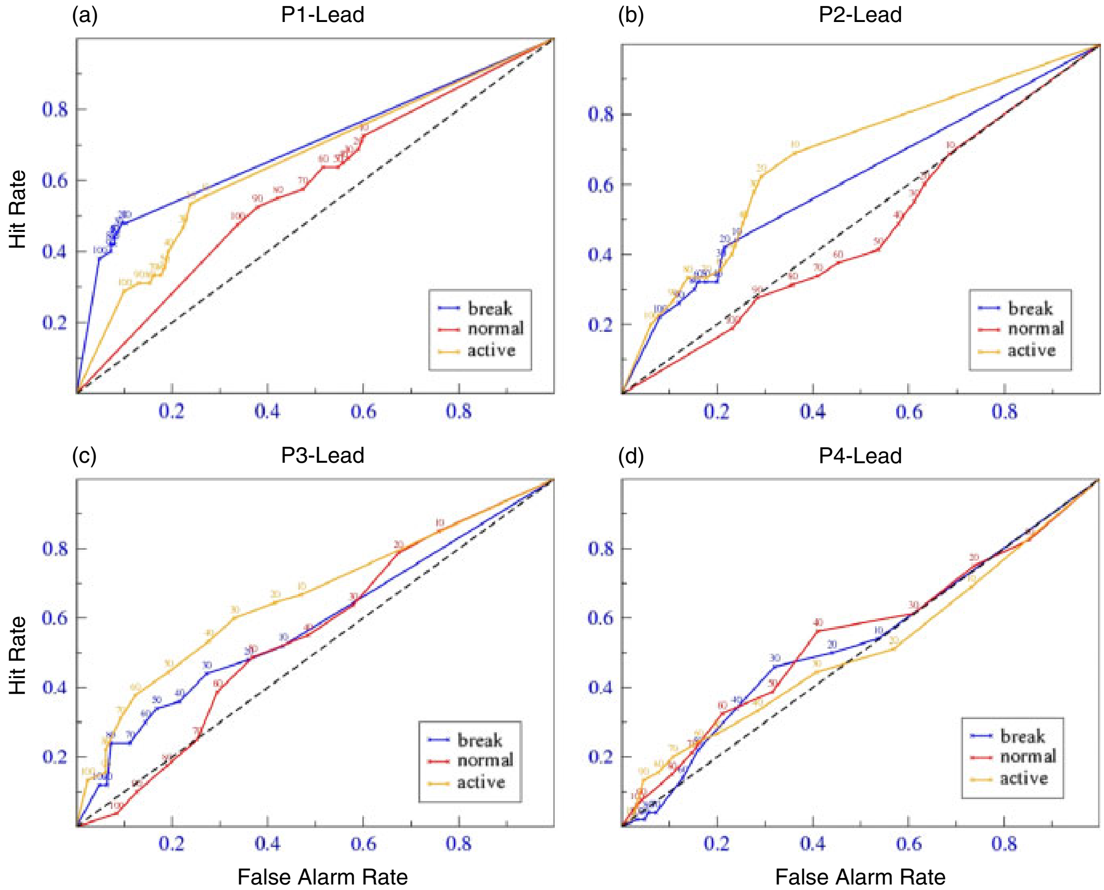

Figure 12. The ROC for three categories of rainfall (break, normal and active) averaged over MZI region and the probability values are also marked in the figure.

Main advantage of the EPS developed in the CFSv1 model is that it is a reliable and flexible system, which is simple to implement in operational environment and capable of generating as many members, if provided adequate computer resources. The impact of the magnitude of high and low perturbation is tested and found that there exists slightly higher dispersion for HIGHPER experiment. Impact of humidity perturbation is also tested and found that HIGHER-NQ experiment has a slightly lower dispersion than HIGHPER experiment (see Figure 2).

The climatological seasonal cycle in precipitation is well captured by the EPS system and the northward propagation is also well simulated. The robustness of the system is evaluated by computing the spread, signal and RMSE of the variables. It is found that the spread is always less than RMSE and present EPS is not capable of accounting all possible sources of errors from both initial conditions as well as model error. Vialard *et al.* (2003) pointed out that the underestimation of the initial uncertainty may come from the unpredictable component of atmospheric forcing leading to an underestimation of ensemble spread. Model error may also contribute significantly to the underestimation of ensemble spread. Model error can be reduced by improving the model physics and dynamics. The upper bound of predictability is about 20–25 d for zonal wind as measured by growth of predictability error and lower bound of practical predictability as measured by forecast error is around 10–12 d. The coherent evolution of the spatial structure of spread, signal and RMSE suggests that evolution of noise provides a prior information about the unpredictability in the forecast.

An attempt has also been made in this study to investigate the potential predictability limit of precipitation and other dynamical variables by computing the SNR. We found predictability limit of 15 d (18 d) for precipitation (500 hPa geopotential height) over Indian monsoon region using unfiltered raw model forecasts (see Figure 9). The limit on predictability from this study is comparable to the limit suggested by empirical methods of ISO prediction based on model or observational data (Waliser *et al.*, 1999). As we analysed unfiltered data, one-to-one comparison with earlier studies by Waliser *et al.* (2003a) and Liess *et al.* (2005) is not done because they used either filtered data or model data projected to four leading EOF's. This analysis can be extended to quantify the potential predictability limit during different phases of ISO's from dynamical forecasts of AOGCM.

The skill of predicting precipitation over MZI region suggests that EPS has little or no skill beyond third pentad lead as ACC values dropped below zero after pentad 3.

However, dynamical variables show higher ACC than precipitation and reaches near zero value only after fifth pentad (see Figure 11). The area under the ROC curves for different probabilities is analysed for three categories of precipitation namely, break, active and normal (see Figure 12). Skill of predicting break and active is slightly higher than that of normal. Unlike pure dynamical EPS, some value-added useful probabilistic forecasts can be generated even beyond pentad 3 lead by utilizing the ensembles.

Preliminary result show positive impact of EPS on the extended range prediction of active-break cycle of ISMR. The same procedure will be extended with different combinations of physical parameterizations. Two sets of experiments with two different cumulus parameterization schemes, namely Simplified Arakawa Schubert (SAS) and Relaxed Arakawa Schubert (RAS) schemes will be conducted. Each set of experiments will be run with same 10 perturbed initial conditions by keeping same model configuration and physical parameterization but with two different cumulus schemes and also with stochastic physics perturbation. Future work will also investigate the impact of the resolution in CFS version 2 and number of ensemble members on the EPS prediction skill, impact of various ensemble creation techniques on the design of EPS and finally the development of different types of hybrid forecasting strategies including probabilistic forecast based on EPS. The forecast skill will be further evaluated by studying relevant dynamical and moist convective features associated with ISO.

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